www.tntextbooks.in

example, the ostrich, emu and penguin are all birds but cannot fly. So Aristotle would not have classified them as birds. In spite of these limitations Aristotle's classification system was followed for more than 2000 years upto 1700.

After Aristotle, his student **Theophrastus** (372-287 BC) continued his research on the classification of plants, and he was known as the "Father of Botany." There was a huge gap till 16th century, then the English naturalist **John Ray** (1627–1705) wrote several important works through his life. His most important contribution was the establishment of species as the ultimate unit of taxonomy. In 1682 he published the *Methodus Plantarum Nova*, which contained about **18,000** plant species, a result of a relatively narrow species concept. His complicated classification was based on many combined characters, as opposed to earlier taxonomists. John Ray also aimed at publishing a complete system of nature, which included works on mammals, reptiles, birds, fishes and insects. The Swedish biologist **Carolus Linnaeus (1707 - 1788)** father of modern taxonomy and founder of modern systematics developed a scientific system of taxonomy and binomial nomenclature, which is still (with modifications) in use.

Aristotle to Linnaeus employed easily observable single to few traits for classification of organisms. With increased knowledge of the several biological domains, many characters were considered for classifying organisms. This represented the phase of classical taxonomy which was based on overall similarities or affinities derived from morphology, anatomy and embryology of organisms. A modification of this system is the numerical taxonomy, which evolved in the 1950s. This system evaluates the resemblances and differences through statistical methods followed by computer analyses to establish the numerical degree of relationship among individuals. Later on biologists initiated studies on the evolutionary and genetic relationships among organisms, which led to the emerge of **phylogenetic classification or cladistics.** It is an evolutionary classification based on how a common ancestry was shared. Cladistic classification summarizes the genetic differences between all species in the 'phylogenetic tree'. Ernst Haeckal introduced the method of representing evolutionary relationships with the help of a tree diagram known as cladogram.

This system of classification takes into account ancestral characters (traits of basic body design which would be in the entire group) and derived characters (traits whose structure and functions differs from those of ancestral characters). One or more derived characters which appeared during evolution resulted in the formation of new subspecies. In a cladogram each evolutionary step produces a branching and all the

Figure 1.1. Example of a Cladogram

